

Quantum Gravitation

Problem Set 10

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Topic

The 3.5 K Thermal Medium

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in The 3.5 K Thermal Medium. State all assumptions and conventions.

$$\rho_T = Z^{-1} e^{-H/k_B T_m}$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$n_B(\omega) = 1/(e^{\hbar\omega/k_B T_m} - 1)$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.

$$T_m = 3.5K$$

Show enough intermediate work that another student can reproduce the calculation.